SENTIMENTRI V2.0 • SECURITY & AUDIT

Audience Quality & Authenticity Guide

Client reference outlining the 5-Signal scoring system, privacy controls, and audit examples.

> In modern influencer marketing, follower counts alone are a highly unreliable metric. Bot farms, comment pods, and fake follower packages artificially inflate creator profiles. Sentimentri's **Audience Authenticity Engine** runs a rigorous multi-platform audit using five distinct engagement and velocity signals to output a single **Audience Authenticity Score (0-100%)**.

1. The 5-Signal Scoring Engine

Rather than relying on basic follower ratios, Sentimentri parses post-level data across all connected profiles to check for automated patterns.

SIGNAL S1

Platform-Specific Engagement Rate (ER) Anomaly (25% Weight)

Engagement benchmarks vary significantly by platform. A TikTok account naturally commands a higher ER than an X account. Using flat thresholds leads to false positives. We evaluate ER based on platform-specific ranges:

Platform	Healthy ER Range	Suspiciously Low	Suspiciously High
Instagram (IG)	1.0% – 5.0%	< 0.5% (ghost follower risk)	> 12.0% (bot farm engagement)
TikTok (TT)	3.0% – 10.0%	< 1.0% (purchased accounts)	> 20.0% (automated loops)
Twitter/X (X)	0.3% – 2.0%	< 0.1% (inactivity)	> 8.0% (repost pods)

SIGNAL S2

Likes-to-Comments Ratio (LCR) Audit (20% Weight)

Genuine human followers discuss and ask questions. Bots usually only click "like" because posting comments requires more advanced APIs. Extremely high LCR ratios are highly indicative of paid likes. Conversely, extremely low LCR values signify comment pods.

```
Likes-to-Comments Ratio (LCR) = Average Likes / Average Comments
```

LCR Range	Authenticity Score	Diagnosis
5 – 100	90 – 100	HEALTHY Balanced organic interaction
100 – 200	60 – 89	SLIGHTLY UNBALANCED Typical of viral media posts
200 – 500	20 – 59	SUSPICIOUS Potential paid like packages
> 500 or 0 Comments	0 – 19	CRITICAL High likelihood of automated bot likes
< 2 (with comments > 10)	30 – 50	POD WARNING Engagement group or chat pods spamming comments

SIGNAL S3

Comment Quality NLP Signal (25% Weight)

We analyze comment text from the last 10 organic posts. We use natural language processing (NLP) to classify comments. Emoji-only replies ("□□□") or generic remarks ("Nice post!", "Collab?") are flagged as non-substantive.

```
Substantive Ratio = (Substantive Comments / Total Comments) * 100
```

- **Substantive Ratio > 60%:** Score 90-100 (Highly communicative audience)
- **Substantive Ratio 40% - 60%:** Score 70-89 (Average conversation patterns)
- **Substantive Ratio < 20%:** Score 0-39 (Spam/emoji bots dominant)
- **Engagement Pod Flags > 30%:** Deducts 20 points instantly (high frequency of pod comment patterns detected).

SIGNAL S4

Views-to-Followers Activity Index (15% Weight)

A creator with 1,000,000 followers whose videos average only 2,000 views is exhibiting clear signs of purchased/dormant followers. Real creators always retain a core active reach percentage.

```
Views-to-Followers Ratio (VFR) = (Average Views / Followers) * 100
```

- **VFR > 20%:** Score 90-100 (Extremely active audience pool)
- **VFR 5% - 10%:** Score 50-69 (Average active reach index)
- **VFR < 1%:** Score 0-19 (Ghost / purchased account indicators)

SIGNAL S5

Engagement Consistency Variance (15% Weight)

Organic engagement is naturally noisy, fluctuating based on content topic, time of day, and algorithm performance. Bot engagement packages are usually purchased for specific posts, leading to extreme, unnatural spikes. We calculate the Coefficient of Variation (CV) across organic benchmarks.

```
CV = Standard Deviation of Engagement / Mean Engagement
```

- **CV < 0.3:** Score 90-100 (Consistent community interaction)
- **CV 0.3 - 0.6:** Score 70-89 (Healthy content-driven variation)
- **CV > 1.0:** Score 0-39 (Extreme, highly irregular spikes indicating bot runs)

2. Client Data Safety & Sensitivity Toggle

Audience authenticity scores are highly sensitive. Falsely accusing an organic creator of having fake followers can damage relations, while failing to spot a bot farm wastes client budgets.

The Visibility Gate Control

To manage this risk, Sentimentri provides a workspace-wide privacy toggle: **"Show Sensitive Scores"**.

- **Visibility OFF (Default):** Authenticity scores and signals are blurred in the dashboard. Roster cards show **"Score Hidden "** in neutral grey. This prevents clients or viewers from seeing unverified ratings.
- **Visibility ON:** Authenticity scores, signal breakdowns, platform sub-scores, and warning alerts are revealed to authorized team members.

3. Creator Audit Case Studies

Below are three representative creator audits illustrating the engine's capability to dissect audience profiles.

Case A: The Organic Creator (Health: Excellent)

94% AUTHENTIC

Creator Profile: Lifestyle influencer, 120,000 followers.

Signal	Value	Score	Analysis
S1: ER Anomaly	3.4% (IG)	95/100	Well within the healthy 1.0%-5.0% Instagram benchmark.
S2: LCR Ratio	35.2	95/100	Healthy interaction distribution (~35 likes per comment).
S3: NLP Quality	72% Substantive	95/100	Comments contain context-rich discussions; no pod flags.
S4: Views Ratio	22.1%	95/100	Excellent active viewership ratio.
S5: CV Variance	0.42	80/100	Normal fluctuation based on video topic performance.

Recommendation: Approved for high-fee campaign activations. Highly engaged, authentic community.

Case B: The Purchased Follower Profile (Health: Critical Bot Risk)

28% AUTHENTIC

Creator Profile: Tech reviewer, 650,000 followers.

Signal	Value	Score	Analysis
S1: ER Anomaly	0.12% (IG)	25/100	CRITICAL ER is far below the 0.5% threshold for 10K+ accounts.
S2: LCR Ratio	780.0	10/100	CRITICAL Disproportionate likes vs comments. Likelihood of bot farm.
S3: NLP Quality	8% Substantive	20/100	92% of comment feed is emoji-only or automated spam.
S4: Views Ratio	0.45%	10/100	Huge follower count with near-zero video views.
S5: CV Variance	1.25	20/100	Unnatural, massive engagement spikes only on sponsored posts.

Risk Alert: □ CRITICAL: Extreme ghost follower presence and inorganic likes detected. Do not activate.

Case C: The Comment Pod Participant (Health: Warning) 43% AUTHENTIC

Creator Profile: Fashion blogger, 45,000 followers.

Signal	Value	Score	Analysis
S1: ER Anomaly	1.8% (IG)	95/100	ER falls inside normal Instagram thresholds.
S2: LCR Ratio	1.8	40/100	WARNING LCR < 2 indicates abnormal comment volume vs likes.
S3: NLP Quality	35% Substantive	35/100	Comment pod flags triggered. Reciprocal commenting patterns.
S4: Views Ratio	4.2%	35/100	Relatively low viewership for a 45K account.
S5: CV Variance	0.31	80/100	Consistent engagement distribution.

Risk Alert: ⚠ WARNING: Comment pod activity and recurrent reciprocal commenting detected. Proceed with caution.